

Here's a comprehensive test list covering every aspect of your system:

Layer 0 — Whitelist Tests

Should be BENIGN (whitelist) — root domains

<https://google.com>

<https://github.com>

<https://paypal.com>

<https://microsoft.com>

Should be BENIGN (whitelist) — trusted subdomains (FIX 11)

<https://mail.google.com/mail/u/0/>

<https://docs.python.org/3/library/os.html>

<https://api.github.com/repos/user/project>

<https://login.microsoftonline.com/tenant>

<https://accounts.google.com/signin>

<https://www.youtube.com/watch?v=dQw4w9WgXcQ>

<https://support.apple.com/en-us/HT201250>

Should NOT be whitelisted — goes to ML (FIX 11 subdomain attack)

<http://google.com.evil.tk/steal>

<http://paypal.com.verify-now.net/login>

<http://secure.google.com.phishing.ru/login>

<http://login.google.com.evil.xyz/steal>

Layer 1 — Google Safe Browsing Tests

Known phishing URLs — should be MALICIOUS (GSB)

<http://testsafebrowsing.appspot.com/s/phishing.html>

<http://testsafebrowsing.appspot.com/s/malware.html>

Layer 2 — ML Model Tests

Leet Speak (FIX 9)

Should be MALICIOUS 85%+ (leet override)

<http://g00gle.com/login>

<http://paypa1.tk/secure>

<http://amaz0n-login.xyz/verify>

<http://faceb00k.gq/checkpoint>

<http://m1cr0s0ft.top/update>

<http://1nstagram.cf/login>

<http://netfl1x.xyz/signin>

<http://linked1n.com/verify>

<http://wh4tsapp.tk/activate>

<http://p4ypal.com/secure>

Brand Impersonation (no leet)

Should be MALICIOUS

<http://paypal-security-alert.com/verify>

<http://amazon-verify-account.net/login>

<http://microsoft-support-center.com/fix>

<http://apple-id-locked.com/unlock>

<http://google-security-alert.tk/verify>

<http://paypal.suspicious-login.com/verify>

Phishing Patterns

Should be MALICIOUS

<http://secure-banking-update.tk/account/verify.php>

<http://free-iphone-winner.xyz/claim/prize/now>

<http://account-suspended-verify.ml/login>

<http://password-reset-required.ga/update>

<http://xn--pypal-4ve.com/signin>

<http://192.168.1.1/paypal/login>

<http://185.220.101.45/malware/payload.exe>

IP Based

Should be MALICIOUS

<http://192.168.1.1/login/secure/account>

<http://185.220.101.45/steal/credentials>

<http://10.0.0.1/banking/verify>

Suspicious File Extensions

Should be MALICIOUS

<http://microsoft-alert.support/windows/virus-detected.exe>

<http://free-download.tk/setup.scr>

<http://update-required.ml/patch.msi>

URL Shorteners

Verdict depends on destination

<http://bit.ly/3xK9mNp>

<http://tinyurl.com/suspicious>

Layer 3 — Domain Age Tests

Newly registered domains — should get age boost (×1.4)

(test any URL from a domain registered < 30 days ago)

Old established domains — should get age reduction ($\times 0.3$)

These should be BENIGN even if ML says borderline

<https://stackoverflow.com/questions/12345/how-to-code>

<https://wikipedia.org/wiki/Phishing>

Edge Cases

Legitimate Sites With Many Dots (FIX 11 Problem 2)

Should be BENIGN — go through ML and pass

<https://docs.djangoproject.com/en/4.2/topics/security/>

<https://developer.mozilla.org/en-US/docs/Web/API>

<https://aws.amazon.com/ec2/pricing/>

<https://cloud.google.com/storage/docs>

<https://dev.twitter.com/rest/public>

Legitimate HTTP Sites (no HTTPS)

Should be BENIGN — old HTTP sites without suspicious signals

<http://neverssl.com>

<http://httpforever.com>

<http://example.com>

Long Legitimate URLs

Should be BENIGN

https://www.amazon.co.uk/dp/B08N5WRWNW?ref=ppx_yo_dt_b_asin_title

<https://github.com/user/repo/blob/main/src/components/App.jsx>

<https://stackoverflow.com/questions/12345678/how-to-fix-python-error>

Tricky Benign

Should be BENIGN — legitimate security sites

<https://haveibeenpwned.com>

<https://virustotal.com>

<https://shodan.io>

<https://portswigger.net/web-security>

Homoglyph / Punycode

Should be MALICIOUS

<http://xn--pypal-4ve.com/signin>

<http://xn--googl-fsa.com>

<http://paypal.com/login> (Cyrillic 'a')

Full Expected Results Summary

Category	Count	Expected
Whitelist root domains	4	BENIGN (whitelist)
Whitelist subdomains	7	BENIGN (whitelist)
Whitelist bypass attacks	4	MALICIOUS (ML)
GSB known threats	2	MALICIOUS (GSB)
Leet speak	10	MALICIOUS 85%+
Brand impersonation	6	MALICIOUS
Phishing patterns	7	MALICIOUS
IP based	3	MALICIOUS
Suspicious extensions	3	MALICIOUS
Legit many dots	5	BENIGN
Legit HTTP	3	BENIGN
Long legit URLs	3	BENIGN
Tricky benign	4	BENIGN

Category	Count Expected	
Homoglyph/punycodes	3	MALICIOUS